NEXVIA AEGIS V9-M4 Methodological Audit

Final Frozen-Policy Provenance and Hypothesis H3 Decision-Method Audit

Autonomous Verification Subsystem (Google DeepMind Antigravity)
Git Branch: v9.0-runtime-assurance · Reported Prior Commit: f3ede86 · Date: October 25, 2026

CRITICAL NON-AUTHORIZATION DIRECTIVE:
This audit document resolves two specific pre-execution questions: (1) Does the production evaluator load and verify the exact frozen policy certified by Milestone M2R2? (2) Does the Hypothesis H_3 decision rule use the confidence-bound method required by the authoritative registered experiment protocol? **This document does NOT constitute authorization to execute Milestone V9-M4.** All evaluations in this report were conducted strictly using synthetic test fixtures, in-memory models, and read-only checks of existing artifacts. Production holdout manifests (`test_mix_manifest_r1.json` and `test_scen_manifest_r1.json`) and holdout output directories (`results/v9/test_mix/` and `results/v9/test_scen/`) do not exist. Reserved holdout seed blocks (73000..74439 for mixture holdout and 75000..75119 for scenario holdout) remain completely uninstantiated and untouched. Live execution requires explicit authorization and delivery of an authorization token from the project owner.

Abstract

This audit resolves two foundational questions governing the pre-execution integrity of Milestone V9-M4: (1) exact provenance and verification of the frozen policy artifact certified by Calibration Round 2 (M2R2), and (2) the authoritative confidence-bound decision method for Hypothesis H_3 (Operational Utility & True-Success Rate Noninferiority).

First, we resolve a clerical citation discrepancy in Report 18, which mistakenly cited the historical calibration artifact `results/v9/cal/selected_policy.json` (Candidate `V9_CAND_2_TIER2_VERY_STRICT`, raw SHA-256 `1c316d2d...`). We reconcile and prove that the authoritative M2R2 certified policy evaluated in Milestone V9-M3 is located strictly at `results/v9/cal_r2/selected_policy.json` (Candidate `V9_CAND_1_TIER1_STRICTEST`, raw SHA-256 `5a89d6061b379f1b34ab47be0deef8d63180d3e4c21d18c478bc7e451f5c84e2`, canonical payload SHA-256 `90dc5c0068bb62187415bba57fbf9cafc1f18e111ad5790b0214961490902034`). The production evaluator `experiments/v9_evaluate_holdout.py` defaults to this path, enforces candidate ID matching fail-closed, and strictly rejects the historical R1 artifact.

Second, we audit the authoritative decision method for Hypothesis H_3. In the preregistered protocol (`docs/v9_experiment_protocol.md` Section 3.2.4 & Table 3.5.1), statistical power and sample size were derived from the paired Wald standard error with Student's t critical value. In prior commit `f3ede86`, an ad-hoc dual-conjunction rule (`Student_t and Newcombe`) was inadvertently implemented. A systematic grid search of 22,478 boundary configurations on $N = 1,200$ reveals that Student's t Wald LCB and Newcombe Method 10 score LCB disagree in 250 configurations (1.11%), demonstrating that they are distinct mathematical operators. We restore strict protocol fidelity in `experiments/v9_evaluate_holdout.py` by anchoring the confirmatory pass/fail decision strictly to the prespecified Student's t Wald LCB (`lcb95_delta_tsr > -0.05` and $\widehat{\mathrm{TSR}}_{\mathrm{nominal}} \geq 0.94$), reporting Newcombe Method 10 as a prespecified sensitivity analysis with an explicit concurrence indicator (`sensitivity_newcombe_concurs`).

Three synthetic regression tests (`test_60`, `test_61`, `test_62`) verify policy provenance, authoritative decision rule behavior, and boundary envelope divergence. All 62 preflight tests and all 173 full repository tests pass with zero failures. Zero real holdout execution occurred.

Keywords: Frozen Policy Provenance, M2R2 Certification, Hypothesis H3, Paired Wald Confidence Bound, Student's t LCB, Newcombe Paired Score Interval, Decision Disentanglement, Common Random Numbers.

1 Executive Summary

1.1 Executive Summary (English)

1. **Frozen Policy Provenance Discrepancy Resolved:** Report 18 contained a clerical error citing `results/v9/cal/selected_policy.json` (`1c316d2d...`). We verify that the authoritative M2R2 certified policy is `results/v9/cal_r2/selected_policy.json` (`5a89d606...`, `V9_CAND_1_TIER1_STRICTEST`). The historical `results/v9/cal/` artifact is a retired Round 1 artifact that is fail-closed rejected by the production evaluator.
2. **"Candidate Parameters Hash" Clarified:** The phrase "Candidate Parameters Hash" used informally in project communications corresponds to the full raw file SHA-256 digest (`5a89d606...`) of the M2R2

policy artifact. The production evaluator explicitly verifies both the canonical payload hash (`90dc5c00...`) and candidate ID.

3. **Authoritative H_3 Decision Rule Disentangled:** The preregistered confirmatory decision rule for H_3 is governed strictly by the paired Student's t Wald lower confidence bound ($\mathrm{LCB}_{95} > -0.05$ on S_{10} and $\widehat{\mathrm{TSR}}_{\mathrm{nominal}} \geq 0.94$). Imposing an unprespecified dual hurdle (`Student_t and Newcombe`) was an ad-hoc conservative penalty. We decoupled the rule so that Student's t controls the confirmatory decision, while Newcombe Method 10 is reported as a prespecified sensitivity analysis.
4. **Boundary Concordance Qualification:** The claim of 100% decision concordance between Student's t and Newcombe is bounded to realistic operational distributions and hand-selected benchmark cases. Across the full boundary envelope near -0.05 at $N = 1,200$, 1.11% of contingency tables yield discordant decisions. This mathematical distinction justifies prespecifying the controlling method rather than post-hoc selecting between them.
5. **Common Random Numbers (CRN) Claim Boundaries:** We reaffirm that CRN provides architectural coupling ($t = 0$ bitwise initial conditions and independent stream derivation), but sample variance reduction is an empirical property requiring positive covariance ($\mathrm{Cov}(Y_\pi, Y_{\mathrm{base}}) > 0$). While established on historical benchmark data ($\rho \approx 0.593$), holdout variance reduction remains an empirical outcome to be observed post-authorization.
6. **Synthetic Test Verification:** Added unit tests `test_60`, `test_61`, and `test_62` to `tests/test_v9_holdout_preflig` All 62 preflight tests pass (1.30s); all 173 repository tests pass (27.44s).
7. **Holdout Independence Preserved:** Zero real holdout data or manifests were accessed or created. The frozen policy and calibration artifacts remain untouched.

1.2 บทสรุปสำหรับผู้บริหาร (Executive Summary in Thai)

1. **การพิสูจน์ความถูกต้องของไฟล์นโยบายคงที่ (Frozen Policy):** รายงานฉบับที่ 18 มีข้อผิดพลาดทางธุรการในการอ้างอิงพาธไฟล `results/v9/cal/selected_policy.json` (`1c316d2d...`) การตรวจสอบอย่างละเอียดข้ามเอกสารพิธีสารยืนยันว่า ไฟล์นโยบายที่ได้รับการรับรองอย่างเป็นทางการจากรอบการปรับเทียบ M2R2 คือ `results/v9/cal_r2/selected_policy.js` (`5a89d606...`, ผู้สมัคร `V9_CAND_1_TIER1_STRICTEST`) ในขณะที่ไฟล์ในไดเรกทอรี `results/v9/cal/` เป็นไฟล์ประวัติศาสตร์ R1 ซึ่งระบบประเมินผลจริงถูกเขียนให้ปฏิเสธการโหลดทันที (Fail-Closed)
2. **ความกระจ่างของคำว่า "Candidate Parameters Hash":** คำว่า "Candidate Parameters Hash" ที่ถูกใช้ในการสื่อสารแบบไ แท้จริงแล้วหมายถึงค่า Raw SHA-256 Digest (`5a89d606...`) ของไฟล์นโยบายรอบ M2R2 ทั้งนี้ ตัวประเมินผลในโค้ดการผลิต Canonical Payload Hash (`90dc5c00...`) และ Candidate ID อย่างเคร่งครัด
3. **การแยกแยะเกณฑ์การตัดสินสมมติฐาน H_3 ตามพิธีสารที่ลงทะเบียนไว้:** เกณฑ์การตัดสินยืนยันผล (Confirmatory Decision Rule) สำหรับสมมติฐาน H_3 ถูกกำหนดไว้ในพิธีสารให้ใช้ paired Student's t Wald Lower Confidence Bound ($\mathrm{LCB}_{95} > -0.05$ บนชุดสถานการณ์ S_{10} ร่วมกับเกณฑ์ $\widehat{\mathrm{TSR}}_{\mathrm{nominal}} \geq 0.94$) การนำเกณฑ์เชื่อมด้วย AND (`Student_t and Newcombe`) มาใช้ในรอบก่อนหน้าถือเป็นการเพิ่มอุปสรรคที่ไม่ได้รับการลงทะเบียนไว้ล่วงหน้า เราได้ทำการปรับปรุงโค้ดให้ Student's t เป็นเกณฑ์ชี้ขาดหลักแต่เพียงผู้เดียว และรายงาน Newcombe Method 10 เป็นการวิเคราะห์ความไว (Sensitivity Analysis) พร้อมตัวบ่งชี้ความสอดคล้อง (`sensitivity_newcombe_concurs`)
4. **ข้อจำกัดของความสอดคล้องระหว่างวิธี Student's t และ Newcombe:** ข้ออ้างที่ว่าวิธี Student's t และ Newcombe สอดคล้องกัน 100% นั้น เป็นจริงเฉพาะบนสภาวะการกระจายตัวทั่วไปและ 7 กรณีขอบเขตตัวแทนเท่านั้น แต่จากการค้นหาอย่างละเอียดบนตารางนับ 22,478 แบบ พบความขัดแย้ง 250 แบบ (1.11%) บริเวณขอบวิกฤตใกล้ -0.05 ซึ่งพิสูจน์ว่าทั้งสองวิธีเป็นฟังก์ชันทางคณิตศาสตร์ที่แตกต่างกัน และตอกย้ำความจำเป็นในการยึดถือวิธีที่ระบุไว้ในพิธีสารเป็
5. **การคงขอบเขตข้ออ้างเรื่องการลดความแปรปรวนของ CRN:** คงความชัดเจนว่า CRN เป็นการจับคู่เชิงสถาปัตยกรรม ($t = 0$ Bitwise Match) แต่การลดความแปรปรวนจริงเป็นคุณสมบัติเชิงประจักษ์ที่ต้องการ $\mathrm{Cov} > 0$ ซึ่งบนชุดข้อมูล Holdout จริงยังต้องรอการวัดผลหลังได้รับอนุมัติ
6. **ผลการทดสอบและการควบคุม:** เพิ่มการทดสอบสังเคราะห์ `test_60`, `test_61`, `test_62` โดยชุดทดสอบ Preflight ผ่าน 62/62 รายการ (1.30 วินาที) และชุดทดสอบทั้งระบบผ่าน 173/173 รายการ (27.44 วินาที) ยืนยันว่าไม่มีการแตะต้องชุดข้อมูล Holdout จริงแม้แต่น้อย

2 Phase 1 — Frozen Policy Provenance and Digest Reconciliation

2.1 Policy Artifact Comparison

Table 1 details the properties of the historical Calibration Round 1 artifact versus the authoritative Calibration Round 2 (M2R2) artifact.

Property	**Historical Calibration Artifact (R1)**	**Authoritative M2R2 Certified Policy**
File Path	`results/v9/cal/selected_policy.json`	`results/v9/cal_r2/selected_policy.json`
Candidate ID	`V9_CAND_2_TIER2_VERY_STRICT`	`V9_CAND_1_TIER1_STRICTEST`
Tier Label	Tier 2 (Very Strict)	Tier 1 (Strictest)
Raw File SHA-256	`1c316d2d97ccf1a93f2dcd4c...`	`5a89d6061b379f1b34ab47be0deef8d...`
Canonical Payload SHA-256	`f1ce3df35f3ab2851a716c5e5337...`	`90dc5c0068bb62187415bba57fbf9caf...`
Created UTC	`2026-10-24T14:38:55Z`	`2026-10-25T05:52:52Z`
Calibration Protocol	Protocol Amendment (Superceded)	`docs/v9_m2r2_protocol_amendment.md`
Validation Milestone	Not Evaluated in M3	Certified in Milestone V9-M3
Evaluator Handling	**FAIL-CLOSED REJECTED** (`ValueError`)	**PREFLIGHT_PASS** (Default Target)

Table 1: Forensic reconciliation of calibration policy artifacts.

2.2 Reconciliation and Traceability Findings

- **Authoritative M2R2 Policy:** Milestone M2R2 executed under the registered protocol amendment (`docs/v9_m2r2_protocol_amendment.md`) and certified candidate `V9_CAND_1_TIER1_STRICTEST`. Milestone V9-M3 evaluated this exact policy, confirming an empirical selective risk of $\widehat{R}_{\text{sel}} = 0.0076 \leq 0.05$ on $M_{\text{val}} = 1,189$ accepted episodes (`docs/v9_m3_post_validation_audit.md`).

- **Evaluator Loading Configuration:** In `experiments/v9_evaluate_holdout.py`, line 1859 defaults to `results/v9/cal_r2/selected_policy.json`. Line 81 defines `EXPECTED_SELECTED_CANDIDATE_ID = "V9_CAND_1_TIER1_STRICTEST"`. Loading the historical R1 artifact immediately triggers a fail-closed exception.

- **Correction of Report 18:** Section 7 of Report 18 and manifest `report_manifest_h3_ci_crn_audit.json` erroneously referenced `results/v9/cal/selected_policy.json`. This was a clerical transposition of directory paths. This audit formally supersedes that citation.

- **Candidate Parameters Hash Meaning:** The phrase refers to the raw 256-bit SHA-256 checksum (`5a89d606...`) of the certified policy file. The production harness additionally validates the canonical JSON payload digest (`90dc5c00...`) to guarantee semantic invariability against formatting changes.

3 Phase 2 — Authoritative H3 Decision-Method Audit

3.1 Protocol Specification of Hypothesis H3

In `docs/v9_experiment_protocol.md`, Hypothesis H_3 assesses operational utility and noninferiority of True-Success Rate (TSR) against the primary baseline V8.1 Martingale across the S_{10} scenario domain ($N = 1,200$ matched pairs).

The confirmatory acceptance criteria defined in Table 3.5.1 are:

$$\text{LCB}_{95}(\Delta\text{TSR}) > -0.05 \quad \text{AND} \quad \widehat{\text{TSR}}_{\text{nominal}} \geq 0.94 \tag{1}$$

Section 3.2.4 derives power from the paired Wald standard error:

$$\text{SE}_0 = \sqrt{\frac{q}{N}}, \quad q = \frac{n_{10} + n_{01}}{N} \tag{2}$$

with the 1-sided 95% Student's t critical threshold:

$$\text{LCB}_{\text{Student-}t} = \overline{d} - t_{0.95,N-1}\frac{s_d}{\sqrt{N}} \tag{3}$$

3.2 Decision Rule Decoupling

In commit `f3ede86`, `evaluate_scenario_holdout` was implemented as:

```
h3_v8_passed = bool(h3_v8_lcb95 > -0.05 and h3_v8_newcombe_lcb95 > -0.05 and nominal_tsr_hat >= 0.94)
```

This conjunction imposed an unprespecified dual test. We updated `experiments/v9_evaluate_holdout.py` to anchor the confirmatory pass/fail decision strictly to the registered Student's t Wald LCB:

```
h3_v8_passed = bool(h3_v8_lcb95 > -0.05 and nominal_tsr_hat >= 0.94)
h3_v8_newcombe_concurs = bool(h3_v8_newcombe_lcb95 > -0.05) if not math.isnan(...) else False
```

The production evaluator now reports both metrics transparently without allowing post-hoc method substitution.

4 Phase 3 — Audit of Decision Concordance Claim

4.1 Grid Search Analysis on $N = 1,200$

To evaluate the reported claim of 100% decision concordance between Student's t Wald and Newcombe Method 10, we conducted an exhaustive grid search over all valid boundary configurations where the paired difference $\Delta\text{TSR} \in [-0.07, -0.03]$ on $N = 1,200$:

- Total boundary configurations analyzed: **22,478**
- Concordant configurations (t decision == Newcombe decision): **22,228 (98.89%)**
- Discordant configurations (t decision $\neq$ Newcombe decision): **250 (1.11%)**

4.2 Representative Boundary Discordance

Table 2 shows representative configurations in the razor-thin boundary envelope near -0.05 where the two methods produce divergent decisions.

n_{11}	n_{10}	n_{01}	n_{00}	ΔTSR	Student's t Wald LCB	Newcombe Score LCB	Outcome Discordance
0	0	48	1152	-0.040000	**-0.049316**	**-0.050386**	t PASS / Newcombe FAIL
120	2	50	1028	-0.040000	**-0.049712**	**-0.050149**	t PASS / Newcombe FAIL
200	3	51	946	-0.040000	**-0.049904**	**-0.050150**	t PASS / Newcombe FAIL
800	14	60	326	-0.038333	**-0.049997**	**-0.050036**	t PASS / Newcombe FAIL

Table 2: Boundary configurations exhibiting decision discordance at margin -0.05 ($N = 1,200$).

This proves that Student's t Wald and Newcombe Method 10 are fundamentally distinct mathematical functions. The claim of 100% concordance applies across broad operational regimes, but fails in 1.11% of razor-thin boundary cases, justifying the prespecification of Student's t as the authoritative decision method.

5 Phase 4 — Common Random Numbers (CRN) Claim Boundaries

We reaffirm the four structural and statistical boundaries governing CRN:

1. **Episode-Level Matched Pairing:** Matching occurs strictly at the episode level via identical integer seeds.
2. **Common Initial Conditions:** At $t = 0$, both controllers observe bit-identical initial states and draw from identical disturbance seeds.
3. **Mutable Stream Independence:** Each controller operates on an independent PRNG state. When a controller invokes additional sensor queries, its stream advances, inducing intentional decorrelation for subsequent draws.
4. **Empirical Covariance Requirement:** Variance reduction $\text{Var}(\overline{\Delta}) < \text{Var}_{\text{unpaired}}$ mathematically requires positive covariance $\text{Cov}(Y_\pi, Y_{\text{base}}) > 0$. On historical benchmark data ($N = 500$), empirical correlation was $\rho \approx 0.593$ (yielding 36.2% SE reduction). On unexecuted holdout data, variance reduction remains an empirical property to be measured post-authorization.

6 Phase 5 — Synthetic Verification Suite

We implemented three unit tests in `tests/test_v9_holdout_preflight.py`:

- `test_60_frozen_policy_m2r2_provenance_and_hash_verification`: Validates M2R2 policy raw hash (`5a89d606...`), canonical hash (`90dc5c00...`), and candidate ID (`V9_CAND_1_TIER1_STRICTEST`), asserting fail-closed rejection of the historical R1 artifact.
- `test_61_h3_authoritative_decision_rule_governed_by_student_t`: Validates that `evaluate_scenario_holdout` passes when Student's t Wald LCB exceeds -0.05 while Newcombe score LCB fails, recording `sensitivity_newcombe_c` `= False`.
- `test_62_h3_concordance_boundary_envelope_qualification`: Validates mathematical divergence across the razor-thin boundary envelope on $N = 1,200$.

Test suite execution:

- Focused Preflight Tests: **62 passed** in 1.30s (Exit Code: 0).
- Full Repository Suite: **173 passed** in 27.44s (Exit Code: 0).

7 Artifact Manifest and Verification Status

Table 3 enumerates all key artifacts and SHA-256 digests.

Artifact Description	**Relative File Path**	**SHA-256 Checksum**
M2R2 Certified Policy	`results/v9/cal_r2/selected_policy.json`	`5a89d6061b379f1b34ab47be0deef8d6...`
Historical R1 Policy	`results/v9/cal/selected_policy.json`	`1c316d2d97ccf1a93f2dcd4c4196aefb...`
M2R2 Candidate Summary	`results/v9/cal_r2/cal_candidate_summary.json`	`043d84a7e93ce48ebca30b3543f07a16...`
M2R2 Calibration Report	`results/v9/cal_r2/calibration_report.json`	`ef3eb0a2b5eecff115e8e8ea6fbb2e97...`
M3 Validation Report	`results/v9/val_r1/validation_report.json`	`a84baea0cae9be1c54b6fb7914f17730...`
Audit Report 19 (Markdown)	`docs/reports/.../19_frozen_policy...md`	`29215033c5e8840742f9b1103c8b41ca...`
Audit Report 19 (LaTeX)	`docs/reports/.../19_frozen_policy...tex`	*(Computed upon compile)*
Audit Report 19 (PDF)	`docs/reports/.../19_frozen_policy...pdf`	*(Computed upon compile)*

Table 3: Cryptographic hashes for verified audit artifacts.

8 Final Recommendation and Non-Authorization Directive

8.1 Formal Audit Status

POLICY_H3_AUDIT_CONFIRMED_WITH_QUALIFICATIONS

8.2 Summary of Audit Conclusions

1. **Policy Provenance:** Certified policy identity is confirmed as `V9_CAND_1_TIER1_STRICTEST` at `results/v9/cal_r2/se` (raw SHA-256 `5a89d606...`, canonical payload SHA-256 `90dc5c00...`). Report 18's citation of `cal/selected_policy.` is formally reconciled as a superseded clerical error.
2. **Decision Rule:** The H_3 confirmatory decision rule is strictly governed by the paired Student's t Wald LCB (> -0.05 on S_{10} and $\widehat{\text{TSR}}_{\text{nominal}} \geq 0.94$). Newcombe Method 10 is prespecified as a sensitivity check with reported concurrence.
3. **Concordance Bounded:** Concordance between Student's t and Newcombe is bounded to operational distributions; boundary discordance occurs in 1.11% of tables near -0.05.
4. **CRN Claims Qualified:** CRN variance reduction is established on historical data, but remains an empirical property to be measured on holdout data post-authorization.
5. **Strict Non-Authorization:** This document does NOT authorize execution of Milestone V9-M4. The production evaluator remains safely locked. Zero real holdout data was generated, accessed, or simulated.